Chapter-18 12-04-2024

Higher-Order-Difference Equations

① Second Order Difference Equations

Standard Form: $Y_{t+2} + a_1 Y_{t+1} + a_2 Y_t = c$

Time Path: $Y_t = Y_p + Y_c$

Particular Solution: $\boxed{Y_p}$

Case-I $\left.a_1 + a_2 \neq -1\right\}$ $y = k$

$$\boxed{Y_p = \frac{c}{1 + a_1 + a_2}}$$

Q1: Find the Y_p From

$Y_{t+2} - 3Y_{t+1} + 4Y_t = 6$

$Y_{t+2} + a_1 Y_{t+1} + a_2 Y_t = c$

$a_1 = -3, \; a_2 = 4, \; c = 6$

$\left.\begin{array}{l} a_1 + a_2 \neq -1 \\ -3 + 4 \neq -1 \\ +1 \neq -1 \end{array}\right]$ check if $a_1 + a_2 \neq -1$

$Y_p = \frac{6}{1 - 3 + 4} = \frac{\cancel{6}}{\cancel{2}}$

$\boxed{Y_p = 3}$

Case-II $\left.a_1 + a_2 = -1\right\}$ $y = kt$

$$\boxed{Y_p = \frac{c}{a_1 + 2} t}$$

Q2: Find Y_p if

$Y_{t+2} + Y_{t+1} - 2Y_t = 12$

$a_1 = 1, \; a_2 = -2, \; c = 12$

check if $a_1 + a_2 = -1$

$1 - 2 = -1$

$-1 = -1$

$Y_p = \frac{12}{1+2} t$

$= \frac{12}{3} t$

$\boxed{Y_p = 4t}$

Case-III $\left.a_1 = -2, a_2 = 1\right\}$ $y = kt^2$

$$\boxed{Y_p = \frac{c}{2} t^2}$$

Q3: Find Y_p if

$Y_{t+2} - 2Y_{t+1} + Y_t = c = 18$

$a_1 = -2, \; a_2 = 1$

$Y_p = \frac{c}{2} t^2$

$Y_p = \frac{18}{2} t^2$

$\boxed{Y_p = 9t^2}$